Benchmarking Counterfactual Explanations of Model Change with Exact Ground Truth

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Abstract. Without ground truth one cannot tell which method for explaining a model update sees which change. We contribute a benchmark with exact ground truth and a linear analysis of what five families of explanations can see, among them differential counterfactuals (Δ-CE): the nearest inputs on which exactly one model changes its decision by a margin. Used as a probe, Δ-CE stays silent on null updates of CNNs (no witness on 2 400 unfiltered queries, nor on 800 when it searches through a learned autoencoder; both registered), while counterfactual and attribution differences report a change. With two independent changes, witnesses point to the one near the query (registered). Gradient differences are blind to an offset and see a rotation, as predicted; a pre-registered hypothesis that Δ-CE recovers the changed factor better than they do was rejected. A margin score tells whether decisions moved; the witness adds a verified input showing where.

1 Introduction

An update that improves accuracy can still change individual decisions in unwanted ways [11]; explaining *how* the models differ is a goal of explainable AI under distributional change [3, 12], distinct from explaining a shift of the data [14]. Artelt et al. [1, 2] compare the counterfactuals [8] of one input under each model, or a difference of gradients; Muschalik et al. [4] compare feature importances; disagreement search [5, 6, 7] looks for inputs on which the models disagree. Ground-truth benchmarks exist for explaining one model [15, 16, 17], not a change, so what each family can see goes unchecked.

We ask which kind of change each family can see, and add *differential counterfactuals* (Δ-CE): the closest input on the data manifold at which only f_{new} changes its decision while f_{old} keeps it, both by a margin κ. Δ-CE anchors disagreement search at a query; unlike robust counterfactuals [9, 10], valid for both models, it seeks one valid for exactly one. We use it as a probe that should stay silent when nothing changed and otherwise return an input that can be checked.

Contributions. (i) A margin witness stays silent on null updates, also unfiltered and through a learned decoder, where counterfactual and attribution differences report change, and it localises two independent changes. (ii) A benchmark with exact ground truth: five rendered factors, changes of four shift types, null updates and a 4×4 grid over the rotation and threshold of the boundary. (iii) A linear analysis separating rotation from offset, tested on CNNs

and a ResNet-8 under a protocol fixed in advance, with registered follow-ups; all failures are reported.

2 Differential counterfactuals and what each family sees

Let $D : \mathcal{Z} \to \mathcal{X}$ map a search space onto the data manifold (here an exact renderer from factor space $\mathcal{Z} = [0,1]^5$). For class y the margin is $m(z) = f_y(D(z)) - f_{1-y}(D(z))$; m_o, m_n belong to f_{old}, f_{new}, and $h = m_n - m_o$. A query z_0 is valid if $m_o(z_0) \geq \kappa$ and $m_n(z_0) \geq \kappa$.

Definition 1. With $\mathcal{F}_\kappa^N = \{z : m_n(z) \leq -\kappa,\ m_o(z) \geq \kappa\}$, a Δ-CE of z_0 is $z^\star \in \arg\min_{z \in \mathcal{F}_\kappa^N} \|z - z_0\|$ if $\|z^\star - z_0\| \leq r$, and NOT_FOUND otherwise; swapping the models gives type O. Each $z \in \mathcal{F}_\kappa^N$ is a *witness*: two forward passes verify that the models decide differently, both decisively, which a difference of two counterfactuals does not.

Lemma 1. Every witness has $h(z) \leq -\kappa - m_o(z) \leq -2\kappa$. Hence models whose margins differ by less than 2κ everywhere have no witness. *Proof:* subtract the two constraints of $\mathcal{F}_\kappa^N$.

Corollary 1 (constant logit offset). If f_{new} adds b to the logit of class 1, e.g. the prior correction [13], then $h \equiv \pm b$ and $\nabla h \equiv 0$. Type-N witnesses exist only for the class whose margin b lowers and only if $|b| \geq 2\kappa$, at the points with $\kappa \leq m_o \leq |b| - \kappa$. *Proof:* insert $h \equiv -|b|$ in Lemma 1; no linearity is needed.

Linear case. For linear margins on $\mathbb{R}^p$ with independent w_o, w_n the nearest witness is unique (supplement). Write $\hat{w}_n = \cos\theta\, \hat{w}_o + \sin\theta\, v$ with $v \perp \hat{w}_o$, and let d_n, d_o be the distances from z_0 to $\{m_n = -\kappa\}$ and $\{m_o = \kappa\}$. The gradient difference of [1] is $-(\hat{w}_n - \hat{w}_o)$, counterfactual differencing $u_n - u_o$ (the two minimum-norm counterfactuals), the witness $u^\star$.

Proposition 2. (a) *Offset* ($\theta = 0$): the gradient difference is 0; counterfactual differencing is a multiple of $\hat{w}_o$; a witness exists iff $d_n \leq d_o$ and equals $-d_n \hat{w}_o$. (b) *Rotation* ($0 < \theta < \pi/2$): the gradient difference lies mainly along the new dependence v, the witness only if $\theta > \pi/4$ or $d_o(\sin\theta + \cos\theta) < d_n$; after projecting out $\hat{w}_o$ all three coincide up to scale. *Proof* of (a): both normals equal $\hat{w}_o$; of (b): supplement.

3 Benchmark and protocol

Data and models. Images are 32×32 ellipses rendered from five independent factors in $[0,1]$: aspect (task factor), size, colour, background and horizontal position; labels are $y = \mathbb{1}[\text{aspect} > 0.5]$ with 5% noise; the renderer is exact, so the factor change of any explanation is read without a learned oracle. All methods search the same factor space through the same renderer, so no method has access the others lack. f_{old} is a small CNN (five seeds); f_{new} is f_{old} fine-tuned on 10 000 samples from one condition (Table 1), as an update on shifted data would be; the shift is injected so that its effect is known. A pair enters only if its change is visible in decisions on validation data. For the grid we measure the

Table 1: Injected changes; $k^\star$: ground-truth factor. [a]$\alpha \in \{0, .1, .25, .5\} \times t \in \{.5, .53, .56, .6\}$, 2 seeds; 4 offsets tested. [b]Split D, 8 qualified; [c]split G; both registered.

Change	Shift	Adaptation data	$k^\star$	Pairs
Shortcut	spurious	colour correlated with label (0.99)	colour	3
Subgroup	local concept	labels flipped for size > 0.75	size	3
Threshold	concept	$y = \mathbb{1}[\text{aspect} > 0.6]$	aspect	3
Null	none / covariate	resample / size $\sim U[0.5, 1]$, same rule	—	5 / 3
Grid[a]	concept	$y = \mathbb{1}[\text{aspect} + \alpha(\text{colour} - 0.5) > t]$	aspect / colour	8 / 24
Offsets[b]	concept	grid, $\alpha = 0$, $t \in \{.53, .56, .6\}$, seeds 2–4	aspect	9
Both[c]	spurious + local	Shortcut and Subgroup at once	colour, size	8

Table 2: Recovery@r (found-rate), split A (locked), mean over three pairs (95% CIs within ± 0.035); last row: found-rate on the 8 null pairs.

	Δ-CE	CF-diff	Grad-diff	Imp-diff	Seg-search	No-prox	CF-new
Shortcut	0.38 (0.84)	0.54 (0.93)	**0.91** (1.0)	0.00	0.02 (0.08)	0.35 (0.63)	0.07 (0.93)
Subgroup	0.53 (0.81)	0.32 (0.99)	**0.62** (1.0)	0.00	0.06 (0.10)	0.29 (0.58)	0.30 (1.0)
Threshold	0.47 (0.47)	0.76 (0.90)	0.07 (1.0)	**1.00**	0.07 (0.08)	0.31 (0.40)	0.90 (0.90)
Null	**0.000**	0.994	1.0	1.0	0.000	0.000	0.996

realised rotation $\hat{\alpha}$ and threshold $\hat{t}$ of each f_{new} by a logistic fit to its decisions and analyse at these values.

Metrics and methods. Per pair: $N = 300$ valid test points. Recovery@r is the share of points with a valid explanation within the budget whose largest factor change is $k^\star$ (NOT_FOUND and ties count 0); it uses type-N witnesses only (pre-declared), so a query whose class the update raises counts 0. All counterfactual methods share one solver, the points and the budget $r = 0.54$, set once on validation. We compare Δ-CE ($\kappa = 0.5$) with **CF-diff**, counterfactual differencing [1]; **Grad-diff**, the gradient difference [1]; **Imp-diff**, a difference of permutation importances [4], one vector per pair; **Seg-search**, a line search towards training points, after [5]; **No-prox**, Δ-CE without proximity, the DeepXplore objective [6], to which DiG-IN [7] reduces on an exact manifold; **CF-new**, the counterfactual of f_{new}; and, in Section 4.4, IG- [18] and SHAP-differences [19].

Protocol. Each test split served at most one confirmatory test, after all choices were fixed (splits A–G in Table 3; dates in the supplement). The follow-ups on splits D, F and G were registered before their data existed [20]. Bootstrap intervals resample points within pairs, so Table 3 also counts pairs and the supplement repeats every test with the pair as unit; the rest is descriptive. A digit tier and a label-shift update failed qualification.

Table 3: Tests (95% CIs; †90%). Pairs: pairs on the claimed side / pairs; $^{\|}$point level only (≤ 4 pairs, pair-level $p \geq 0.0625$); the other confirmed rows also hold with the pair as unit (supplement). K: D = follows from a lemma or definition, R = risky prediction. When: pre = before any result, post = after the first (split B), grid = before the grid run, new = new data, reg = registered [20]. Holm families: *, ‡, §. ¶Run after its registered cost rule had excluded it. expl. = exploratory. Proj.: $\hat{w}_o$ projected out.

Claim	Data	Estimate	p	Pairs	Status	K, when
Abstention and where						
No witness on nulls	Null, B; Adult	0/2 400; 0/1 500		8/8; 5/5	confirmed	D, post; new
No witness on unfiltered nulls	Null, D	0/2 400		8/8	confirmed	R, reg
Same, search via VAE decoder	Null, F	0/800		8/8	confirmed	R, reg
No witness on nulls	ResNet-8, D¶; E, $\kappa^{\star}$	180/200; 0/200		0/2; 2/2	not; expl.	R, reg; new
CF-diff (no cut) flags $\lvert b\rvert < 2\kappa$	Logit offsets	0.961 [0.955, 0.967]	$2{\cdot}10^{-4}$ §		confirmed	D, new
Witness localises ($L > 0$)	Both, G	$L = 0.50$	0.004 (sign)	8/8	confirmed	R, reg
Offsets						
Grad-diff < 0.2	Grid, $\alpha = 0$	0.081 [0.066, 0.096]	$4{\cdot}10^{-4}$ ‡	4/4	confirmed$^{\|}$	R, grid
Grad-diff < 0.2	Offsets, D	0.033–0.093	0.004 (sign)	8/8	confirmed	R, reg
Grad-diff offset-free	Grid	-0.098 [-0.143, -0.053]†	0.94 ‡		not conf.	R, grid
Δ-CE $>$ Grad-diff	Threshold, B	$+0.31$ [0.27, 0.35]	$2{\cdot}10^{-4}$ *	3/3	confirmed$^{\|}$	D, post
Rotations						
Grad-diff rises with α	Grid	$+0.57$ [0.53, 0.61]	$4{\cdot}10^{-4}$ ‡	$\rho = 0.89$	confirmed	R, grid
Proj. Δ-CE $\approx$ Grad-diff	Shortc. B; grid	-0.30; -0.35 (CIs ± 0.03)†	1 *; 1 ‡		not conf.	R, post; grid
Normal at $z^{\star}$ halves gap or more	Grid, new	$-0.19 \rightarrow +0.03$ [0.01, 0.05]†	$6{\cdot}10^{-4}$ §	4/4	confirmed$^{\|}$	R, new
Δ-CE versus gradient differences						
Δ-CE $>$ Grad-diff	Subgroup	-0.09 [-0.13, -0.05]		0/3	rejected	R, pre
Tie ($\lvert\Delta\rvert \leq 0.1$)	Shortc., Thresh.	$\lvert\Delta\rvert = 0.17$–$0.53$			not conf.	R, pre

4 Results

4.1 Abstention, detection and where

The witness stays silent when nothing changed. No disagreement search returns a point on a null update (Table 2). On the 8 null pairs, five null updates of an MLP on Adult, and 8 registered null pairs admitted without the original check (close to the witness criterion), no query has a witness, while CF-diff attributes a change to 97.7–100% of points; this includes the updates fine-tuned on covariate-shifted data. Silence does not need the exact renderer: searching through the decoder of an autoencoder trained on the images (registered; 100 queries per pair, on CPU, not GPU as registered), no null query has a witness (0/800; CF-diff 85–94%), while on 6 of 8 offsets witnesses exist for at least half of the lowered-class queries. The margin is what keeps the probe silent: with $\kappa = 0$ witnesses appear on 83% of null queries, and on offsets the witness rate rises from 0.02 to 1.00 with the largest margin change near the query (descriptive).

Silence costs little sensitivity, but a margin score tells *whether* as well. Each score is cut on the nulls to flag none (or 5%); we count flagged queries of the class an offset lowers. For a constant logit offset b, witnesses appear on every such query from $\lvert b\rvert = 1.1$ (Corollary 1), CF-diff flags ≤ 0.002 up to $\lvert b\rvert = 2$, the margin score $\lvert h(z_0)\rvert$ all from $\lvert b\rvert = 0.9$ at 5%. On the 8 registered fine-tuned offsets, at zero false positives, witnesses exist for 0.17–1.00 of these queries, the margin score flags 0.00–0.54 and CF-diff none. The witness beat the margin score by 0.2 in 7 of 8 pairs, not in all as registered, so we claim no advantage for *whether*.

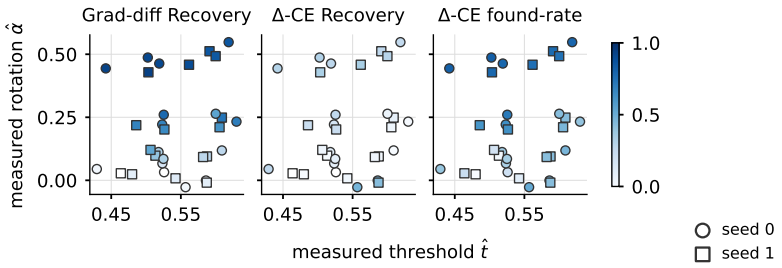


Fig. 1: Grid results at the measured rotation $\hat{\alpha}$ and threshold $\hat{t}$ of each f_{new}; colour scale 0–1; hollow: no $k^\star$; shape: seed of f_{old}.

Witnesses show where. In a registered update with two independent changes, a colour shortcut everywhere and a label flip for large objects, witnesses change size most for 46–75% of queries near the large-object region and for 13–23% far from it (difference $L > 0$ in 8 of 8 pairs, pooled 0.50, sign test $p = 0.004$). Grad-diff and CF-diff barely localise (0.09, 0.13), a global importance difference cannot; the counterfactual of f_{new} alone localises as well (0.51).

4.2 Offsets and rotations

Gradient differences are blind to an offset, as Proposition 2(a) predicts: their Recovery is 0.081 on the pure offsets of the grid and 0.03–0.09 on the 8 registered offsets (8 of 8 below 0.2; sign test $p = 0.004$). On Threshold Δ-CE beats them (0.41 vs. 0.10), in each of 3 pairs, too few for a pair-level test. Type-N witnesses exist only on the side the offset lowers (Threshold: class 1), hence 0.47 in Table 2; type O covers the other side.

Gradient differences see a rotation, increasingly with its angle: Grad-diff Recovery rises from 0.36 at $\alpha = 0.1$ to 0.93 at $\alpha = 0.5$ ($t = 0.5$) (Spearman $\rho = 0.89$ over the 24 pairs, permutation $p = 10^{-4}$; Fig. 1). The projection equivalence of Proposition 2(b) holds only with $\hat{w}_o$ taken at the witness (Table 3).

4.3 Attribution differences, other baselines, a second architecture

With predictions logged beforehand, IG- and SHAP-differences recover colour on Shortcut (0.93) like Grad-diff, and name aspect on Threshold (0.73–0.75 vs. 0.07), against the predicted blindness. The nulls show that this is no signal: there they name aspect on 57% and 71% of queries, and IG-diff on 42% of the registered nulls, as predicted. Imp-diff on test data ranks aspect first in all 9 change pairs, and $k^\star$ on adaptation data. Seg-search with ten times the segments finds witnesses for 0.26–0.47 of queries (predicted ≤ 0.2), none that Δ-CE misses.

A registered ResNet-8 replication shows that κ is model-specific: null fine-tuning moves its margins four times more and witnesses appear on 0.90 of null queries. Calibrating κ on validation nulls (rule fixed in advance; grid extended after $\kappa \leq 2$ failed) gives $\kappa^\star = 2.5$: then no new null query has a witness (0/200),

and witnesses exist for 0.78–1.00 of lowered-class offset queries. Recovery stays 0.36–0.40; Grad-diff is not offset-blind there (0.32–0.35).

5 Discussion and limitations

For *where*, the localisation test is direct evidence, but the counterfactual of f_{new} localises as well; specific to the witness is that it is verified and silent when nothing changed. **Limitations.** (1) The data are synthetic and binary; the learned decoder was tested for abstention and detection only; multi-class tasks, other domains, retraining from scratch and architecture changes are untested; the tabular test covers abstention only. (2) κ is in logit units, exact only for a constant offset, and must be calibrated on null updates of each model family. (3) Witnesses see only decisive changes on the manifold within the budget, without a certificate. (4) Several tests were formed after earlier results; changes were logged before their data existed, with three disclosed exceptions; supplement and code: github.com/Johnlrr/explaining-model-change.

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